Robin Schaefer

Ph.D. Physicist • Solving Quantum Mechanics with High-Performance Computing

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Summary

Computational physicist specializing in high-performance computing and quantum simulation, with experience building scalable scientific software in C++ and Python. Uses computation to understand quantum platforms and quantum materials. Developer of open-source libraries and author of 17 peer-reviewed papers and 8 preprints.

Experience

Postdoctoral Researcher | Harvard University | 2026 – present

- Developing theoretical models and simulation protocols to guide experiments with Rydberg atom arrays and diamond defects
- Engineering an agentic framework for reliable, autonomous research on strongly correlated quantum many-body systems

Postdoctoral Researcher | Helmholtz-Zentrum Berlin | 2025 – 2026

- Modeled magnetic behavior in newly synthesized quantum materials, comparing predictions with experiments
- Implemented a GPU-accelerated quantum Monte Carlo code with a **1,000x** speedup

Postdoctoral Researcher | Boston University | 2023 – 2025

- Discovered novel topological phases using tensor-network methods
- Built simulation software for open and closed quantum systems on **10,000+** cores

Doctoral Researcher | Max Planck Institute, Dresden | 2018 – 2022

- Developed computational models of rare-earth pyrochlores to interpret experiments
- Enhanced a cluster expansion technique to include **500×** more clusters than before

Education

Ph.D. Physics (Dr. rer. nat.) | TU Dresden | 2018 – 2022
Research at the Max Planck Institute for the Physics of Complex Systems

M.Sc. Mathematics | TU Dortmund | 2016 – 2019

M.Sc. Physics | TU Dortmund | 2016 – 2018

B.Sc. Physics | TU Dortmund | 2013 – 2016

B.Sc. Mathematics | TU Dortmund | 2013 – 2016

Grants

2025 | Walter Benjamin Fellowship, German Research Foundation
Awarded competitive funding for an independently proposed research program on neural quantum states (2.5 years)

2022 | Renewal Proposal, Collaborative Research Center CRC-1143
Contributed to the successful **$14 million** renewal proposal for CRC-1143 (2023–2026)

2016 | Scholarship, German Academic Exchange Service
Funded an exchange year at the Korea Advanced Institute of Science and Technology

Honors

2024 | Otto Hahn Medal, Max Planck Society
Max Planck-wide award for outstanding doctoral research

2024 | Best Doctoral Thesis, Collaborative Research Center CRC-1143
Awarded by the center for the best Ph.D. thesis of the year

2018 | Best M.Sc. Physics Student of the Year
Top graduate of the M.Sc. Physics cohort.

Open-Source Libraries

DanceQ

High-performance C++ library for open and closed quantum systems

- Parallelized with OpenMP and MPI
- Divide-and-conquer, symmetry-aware, matrix-free methods
- Interfaces with PETSc and SLEPc
- Peer-reviewed in *SciPost Physics Codebases*
- Documentation & Repository

PyGFMC

GPU-accelerated Green's function Monte Carlo package for zero temperature

- Python package based on JAX
- **~1,000x** speedup over a multi-core CPU
- Submitted to *SciPost Physics Codebases*
- Documentation & Repository

Mentoring & Teaching

- Mentored 5 graduate students and a summer intern in computational methods, software development, and research planning, leading to 6 publications
- Teaching assistant for 4 university classes

Programming & Tools

C++
Python
Julia
Git

HPC

MPI
OpenMP
GPU computing

ML & AI

JAX
LangGraph
Neural quantum states
Agentic workflows

Methods

Monte Carlo
Numerical linear algebra
Optimization
Statistical analysis
Tensor networks